



Modeling Resolution of Pain After Surgery

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Conflicts of Interest

- No financial relationships with industry
- Statistical Editor/Consultant
 - *Anesthesiology*
 - *Cephalalgia*
 - *Headache*
- United States Department of Defense
 - Project MARCH (Lead Statistician)
- Funding: National Institute of Neurological Disorders and Stroke (NINDS)
 - NS06525701
 - GM113852
 - NS380861

Obligatory:

Pain after surgery, It's important

- Pain is nearly ubiquitous after surgery/injury
- Uncontrolled (high) pain after surgery is a risk factor for the development of chronic pain
- Tens of millions of surgeries, hundreds of millions of lost days of productivity, huge costs (\$)
- Persistent pain after surgery/illness can range from 1% (pain after delivery) to 60% (thoracotomy) 1 year after surgery

Defining the Problem:

What to model?

- Acute pain
 - VAS 0-72 hours after surgery, Analgesic requirements, etc.
- Wound healing
 - Size of wound, pain pressure thresholds
- Time to hospital event
 - Hospital Length of stay, time to movement, uptime
- Functional capacity
 - Flexion, activity levels, timed walk
- Recovery Experience
 - Time to 50% pain, pain trajectories
- Persistent Pain
 - Prevalence of > 0 pain, count of pain days, pain disability ratings

Population:

Who to study?

- Child birth
 - Vaginal delivery
 - Cesarean section
- Orthopedic surgery
 - Knee
 - Hip
- Healthy young people
 - Acute pain studies
- Rats
 - Spinal nerve ligation (Chung)
 - Postoperative incisional model (Brennan)

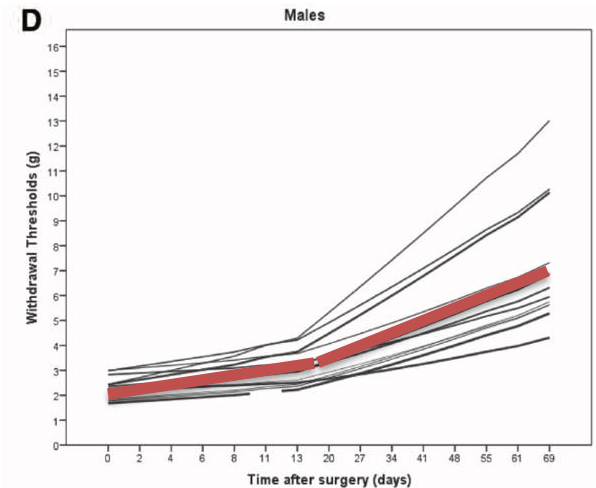
Research Methods:

How to observe pain experience?

- Cross-sectional observational design
 - 0-72 hours
 - 2 weeks
 - 6 months after surgery
- Case-Control
 - Persistent pain matched to no-pain controls
- Longitudinal
 - Cohort
 - Daily-diary
 - Once, twice, EMA

Trajectory Analysis

- (Many) Repeated Measures
- Study of individual change
 - Stability of change
 - Timing/nature of change
 - Heterogeneity across individuals
- Group-level estimates from individual trajectories
 - Can have missing data
 - Can have different measurement occasions



Similar Approaches

Linear-mixed models for longitudinal data

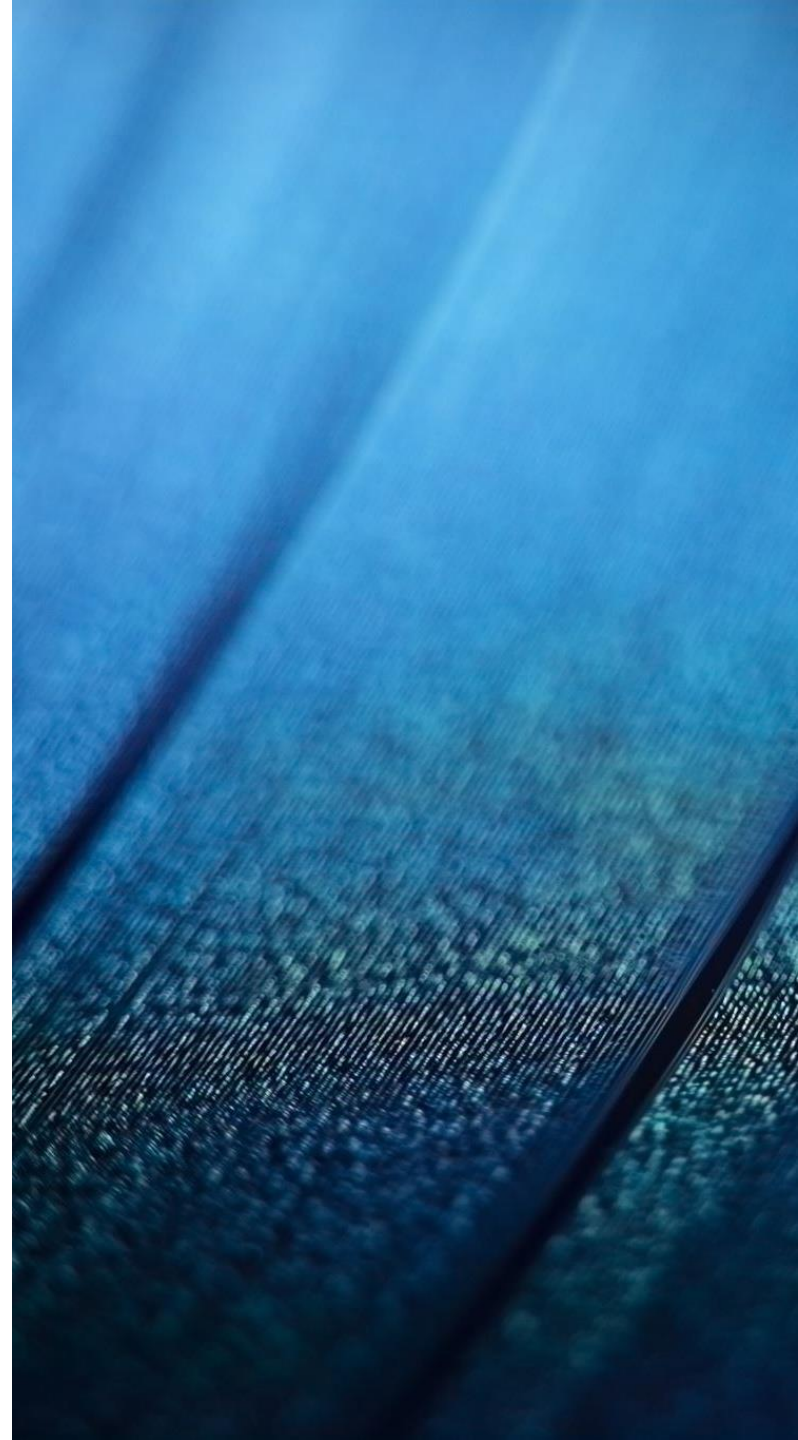
Multilevel models

(Latent) Growth curve analysis

Multilevel polynomial regression

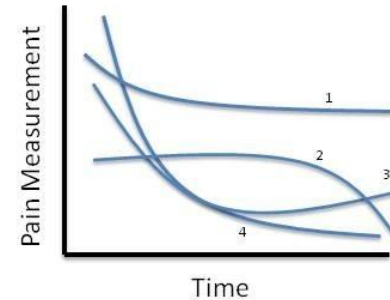
Hierarchical linear models (HLM)

Trajectory analysis



Level 1

$$Pain_i = \pi_{0j} + \pi_{1j}(time) + \pi_{2j}(time^2) + r_{ij}, \text{ where } r_{ij} \sim N(0, \sigma^2)$$



Level 1

Using random effects, a unique pain trajectory is specified for each individual in the study

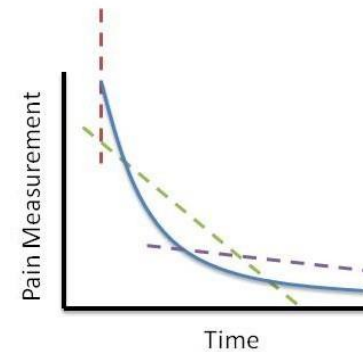
Four individuals are displayed (right) with different pain trajectories

Level 2

$$\pi_{0j} = \beta_{00} + \beta_{01}(\text{Predictor1}) + \beta_{02}(\text{Predictor2})... + u_{0j},$$

$$\pi_{1j} = \beta_{10} + \beta_{11}(\text{Predictor1}) + \beta_{12}(\text{Predictor2})... + u_{1j}, \text{ where } \begin{pmatrix} u_{0j} \\ u_{1j} \\ u_{2j} \end{pmatrix} \sim N \left[\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_{00} & \tau_{01} & \tau_{02} \\ \tau_{10} & \tau_{11} & \tau_{12} \\ \tau_{20} & \tau_{21} & \tau_{22} \end{pmatrix} \right]$$

$$\pi_{2j} = \beta_{20} + \beta_{21}(\text{Predictor2}) + \beta_{22}(\text{Predictor2})... + u_{2j},$$



Level 2

Each parameter in Level 1 is treated as an outcome in level 2

- π_{0j} Variability in starting values across individuals
- π_{1j} Variability in slope across individuals
- π_{2j} Variability in deceleration across individuals

Outline of Studies

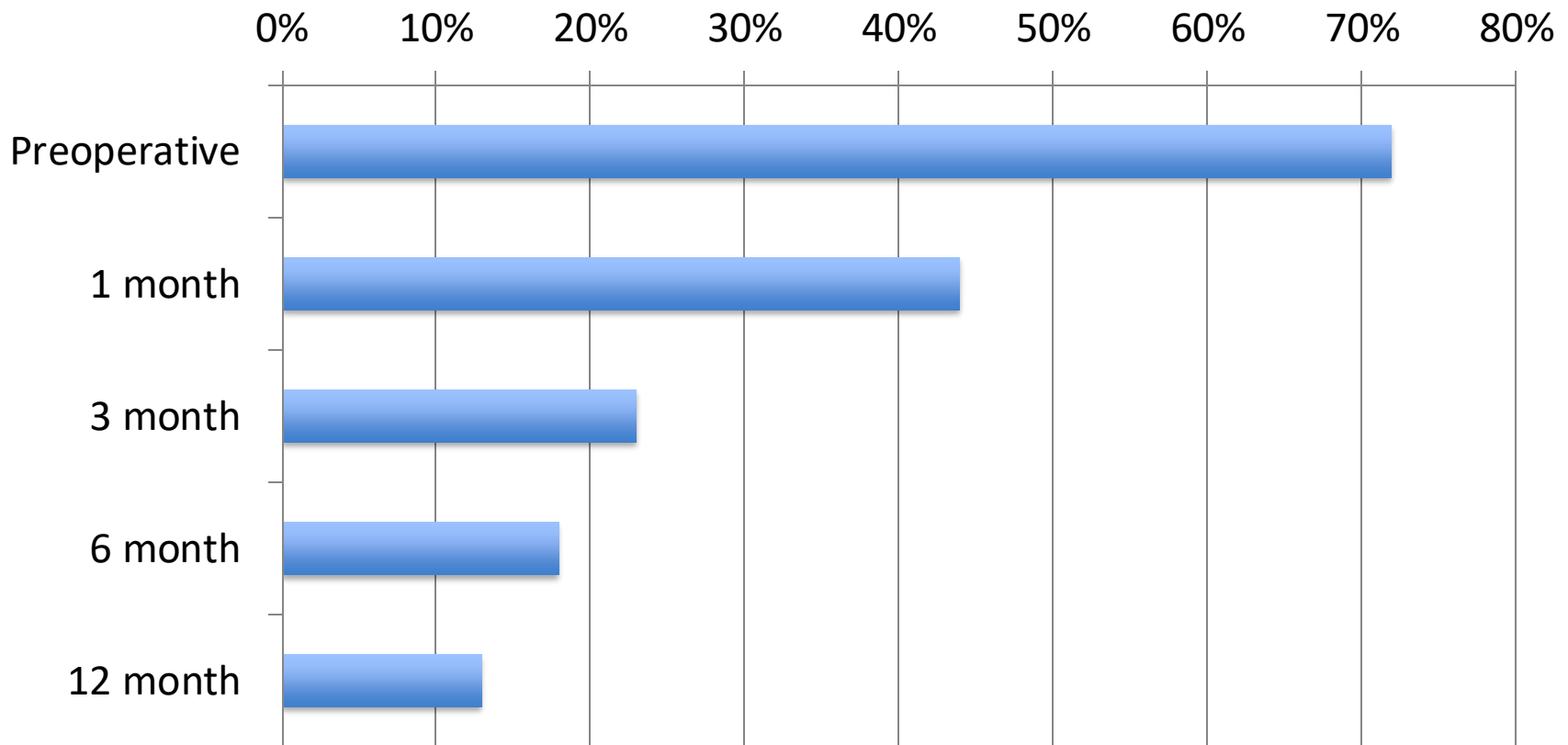
- Predictors of pain after total knee arthroplasty (2002)
- Predictors of pain after cesarean delivery (2006)
- Resolution of pain after childbirth (2012)
- Predictors of acute pain after delivery using 3 questions (2013)
- Day-to-day experiences in recovery after surgery (2017)
- Patterns in recovery after cesarean delivery (2018)
- Recovery of physical activity after cesarean delivery and its relationship with pain (2019)
- Randomized Controlled Trial of intrathecal oxytocin on speed of recovery after hip arthroplasty (2023)

Predicting total knee replacement pain: a prospective, observational study (2002)

[Brander VA1, Stulberg SD, Adams AD, Harden RN, Bruehl S, Stanos SP, Houle T.](#)

N = 116 (149 knees)

Significant pain



Predictors of 12 month pain: preoperative pain, anxiety (STAI), depressive sx's (BDI)

Multifactorial Preoperative Predictors for Postcesarean Section Pain and Analgesic Requirement

Peter H. Pan, M.D.,* Robert Coghill, Ph.D.,† Timothy T. Houle, Ph.D.,‡ Melvin H. Seid, M.D.,§ W. Michael Lindel, M.D.,§ R. Lamar Parker, M.D.,§ Scott A. Washburn, M.D.,§ Lynne Harris, B.S.N.,|| James C. Eisenach, M.D.#

- N = 34 Parturients were enrolled into a psychophysical testing paradigm prior to delivery
- Post-delivery pain and analgesic use was outcome

Table 2. The Six Predictive Factors (Groups) and Their Corresponding Component Predictor Variables

Factor Number	Factor Name	Description
Factor 1	Pain and unpleasantness	This consisted of thermal pain intensity and unpleasantness ratings (measured as VAS of 0–100) when heat stimuli of 49°C applied on the arm and on the back.
Factor 2	Preoperative blood pressure	This consisted of systolic and diastolic blood pressure measured before surgery in the holding area on the day of surgery.
Factor 3	Preexisting pain	This consisted of patients' VAS (0–100) rating of their overall pain and overall unpleasantness during pregnancy.
Factor 4	Expectation	This consisted of patients' VAS (0–100) rating of their expected postcesarean section pain and amount of pain medicine needed.
Factor 5	Thermal pain threshold	This consisted of the threshold temperatures when thermal stimuli applied on the arm and on the back.
Factor 6	Intraoperative factor	This consisted of duration of surgery and the sensory blockade level at time of incision.

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Table 4. Spearman Correlations between Pain Outcomes and Individual Predictor Factors (Groups)

Predictor Factor Variables*	Resting Pain†	Evoked Pain‡	Overall Pain§	Composite Pain Score
1. Pain and unpleasantness	0.45#	0.40**	0.30	0.39
2. Blood pressure	-0.27	-0.09	-0.22	-0.33
3. Preexisting pain	0.13	0.22	0.25	0.21
4. Expectation	0.41**	0.26	0.24	0.34**
5. Thermal pain threshold	-0.036**	-0.42**	-0.14	-0.30
Back threshold (alone)	-0.32	-0.41**	-0.10	-0.37**
6. Intraoperative factor	-0.21	0.02	-0.14	-0.24
Surgical duration (alone)	-0.09	0.13	-0.02	-0.02
7. State Trait Anxiety Inventory	0.30	0.21	0.25	0.31

Table 5. Spearman Correlations between Analgesic Outcome Variables and Individual Predictor Factors (Groups)

Predictor Factor Variables*	Intraoperative†	Recovery Room‡	PCA§	Total Analgesia
1. Pain and unpleasantness	-0.16	0.48#	0.34	0.43**
2. Blood pressure	0.10	-0.02	0.00	-0.01
3. Preexisting pain	0.46	0.11	0.22	0.34
4. Expectation	-0.08	-0.03	0.15	0.14
5. Thermal pain threshold	0.15	-0.43**	-0.05	-0.27
Back threshold (alone)	0.00	-0.41**	-0.08	-0.31
6. Intraoperative factor	0.12	0.04	-0.02	0.01
Surgical duration (alone)	0.01	-0.06	-0.04	-0.02
7. State Trait Anxiety Inventory	0.27	0.39**	0.33	0.40**

Resolution of Pain after Childbirth

2012

James C. Eisenach, M.D.,* Peter Pan, M.D.,† Richard M. Smiley, M.D., Ph.D.,‡
Patricia Lavand'homme, M.D.,§ Ruth Landau, M.D.,|| Timothy T. Houle, Ph.D.#

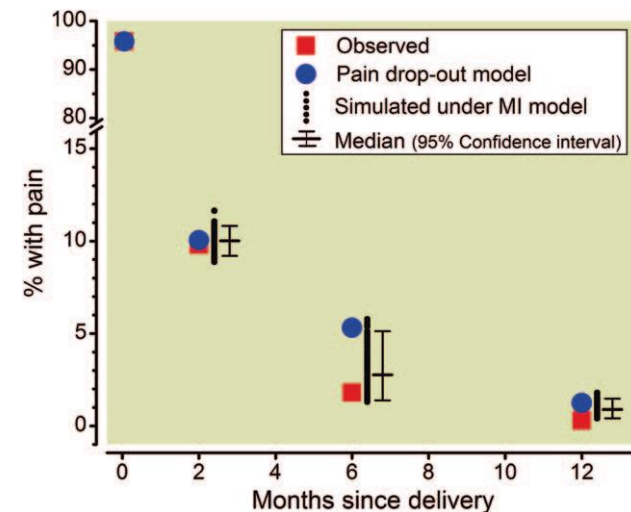
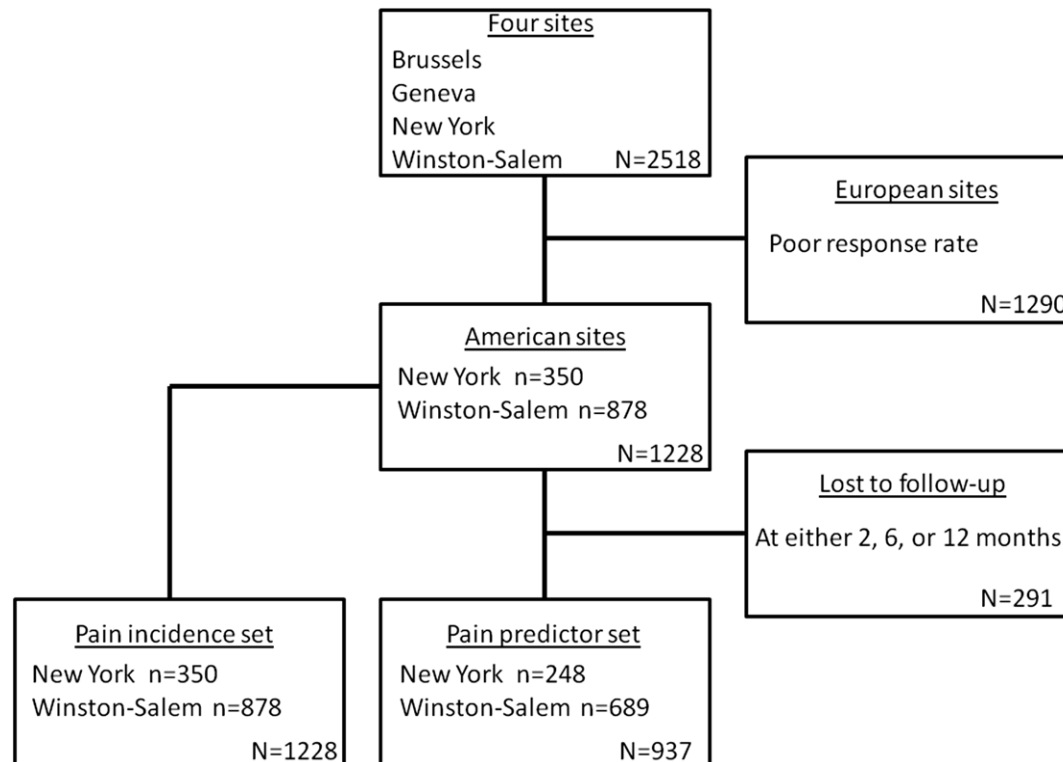


Fig. 2. Pain incidence after delivery, observed or calculated with the pain drop out model. In addition, multiple imputation model results are shown as *small dots* with their median and 95% CI. MI = multiple imputation.

Fig. 1. Participant disposition for both aspects of the data analysis.

Resolution of Pain after Childbirth

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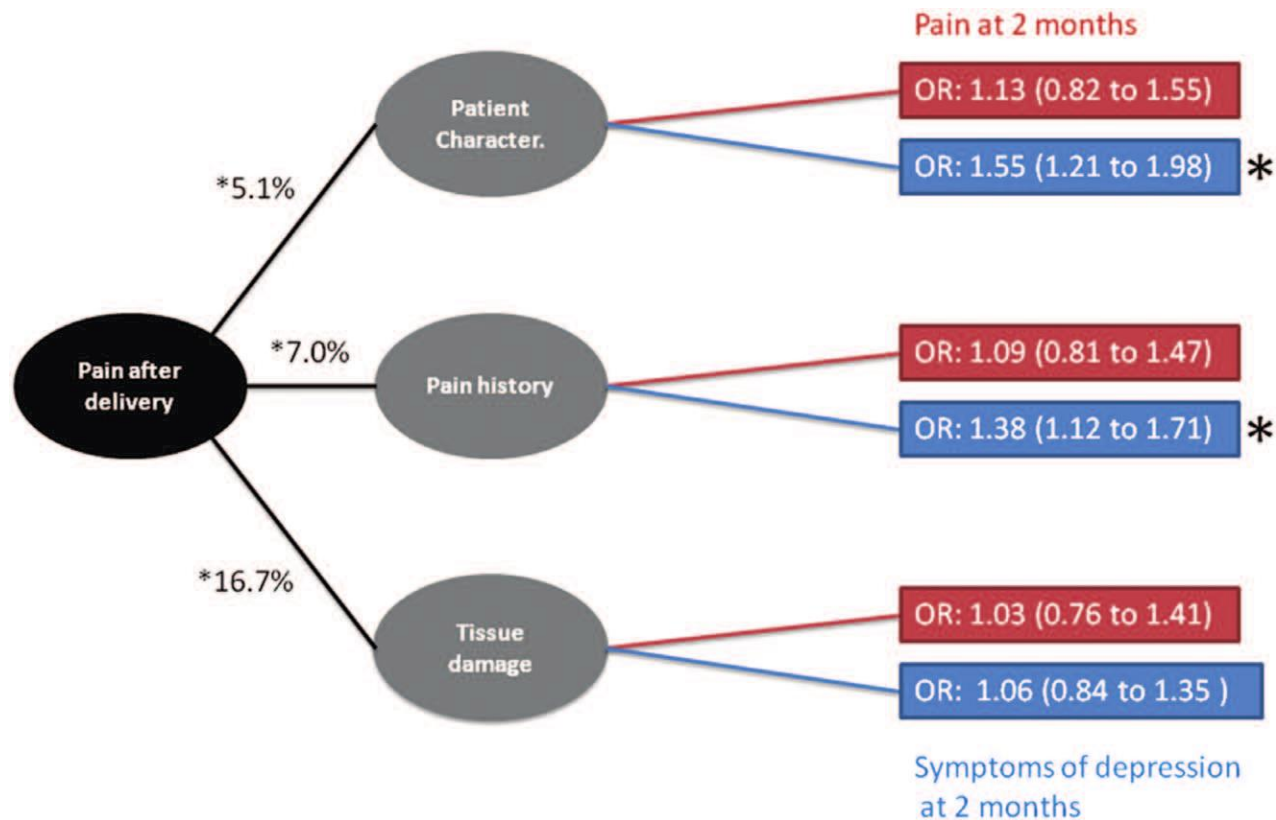


Fig. 3. Predictive model for acute pain after delivery and its application to pain and depression 2 months later. Acute pain after delivery was significantly modeled by each of three factors, patient characteristics, pain history, and tissue damage. These factors failed, however, to result in significant OR for pain at 2 months (red), although patient characteristics and pain history did significantly affect the OR for depression (blue). * $P < 0.05$. OR = odds ratio.

Predicting Acute Pain after Cesarean Delivery Using Three Simple Questions

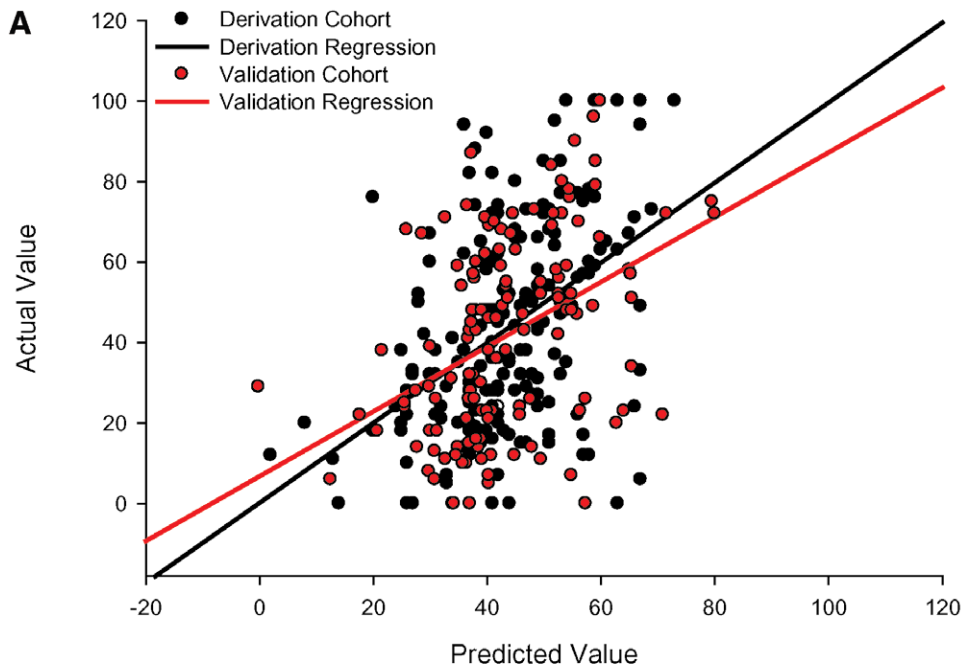
Peter H. Pan, M.S.E.E., M.D.,* Ashley M. Tonidandel, M.D.,† Carol A. Aschenbrenner, M.A.,‡
Timothy T. Houle, Ph.D.,§ Lynne C. Harris, B.S.N.,|| James C. Eisenach, M.D.¶

1. Anxious

2. Anticipation of pain ^A

3. Anticipation of pain
meds

- N = 192 derivation sample
- N = 151 prospective validation sample



Data

**Day-to-day experience in
resolution of pain after
surgery (2017)**

N = 55, N = 157
Total Joint Cesarean

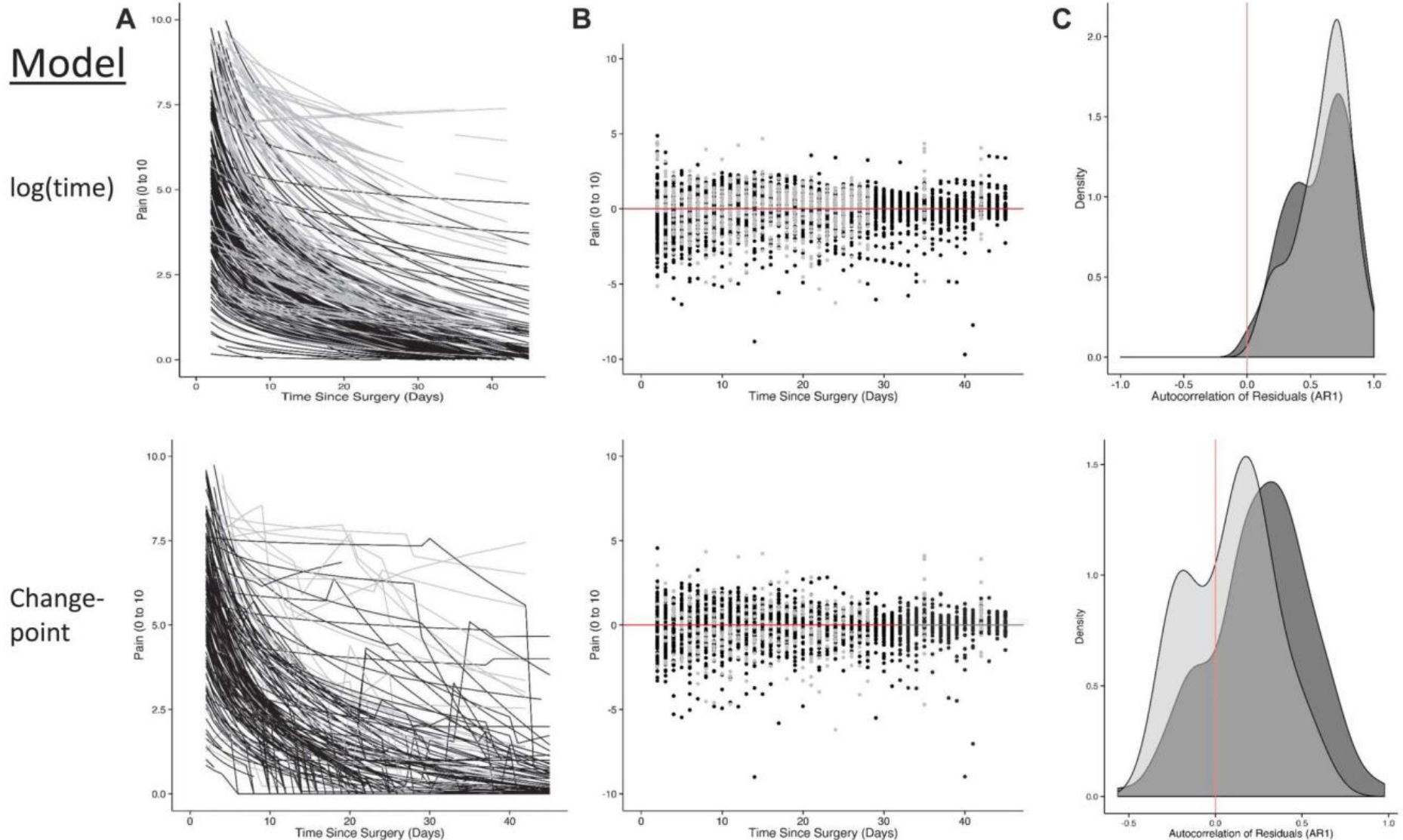
60 days 8 weeks

**Patterns of Recovery from
Pain After Cesarean
Delivery (2018)**

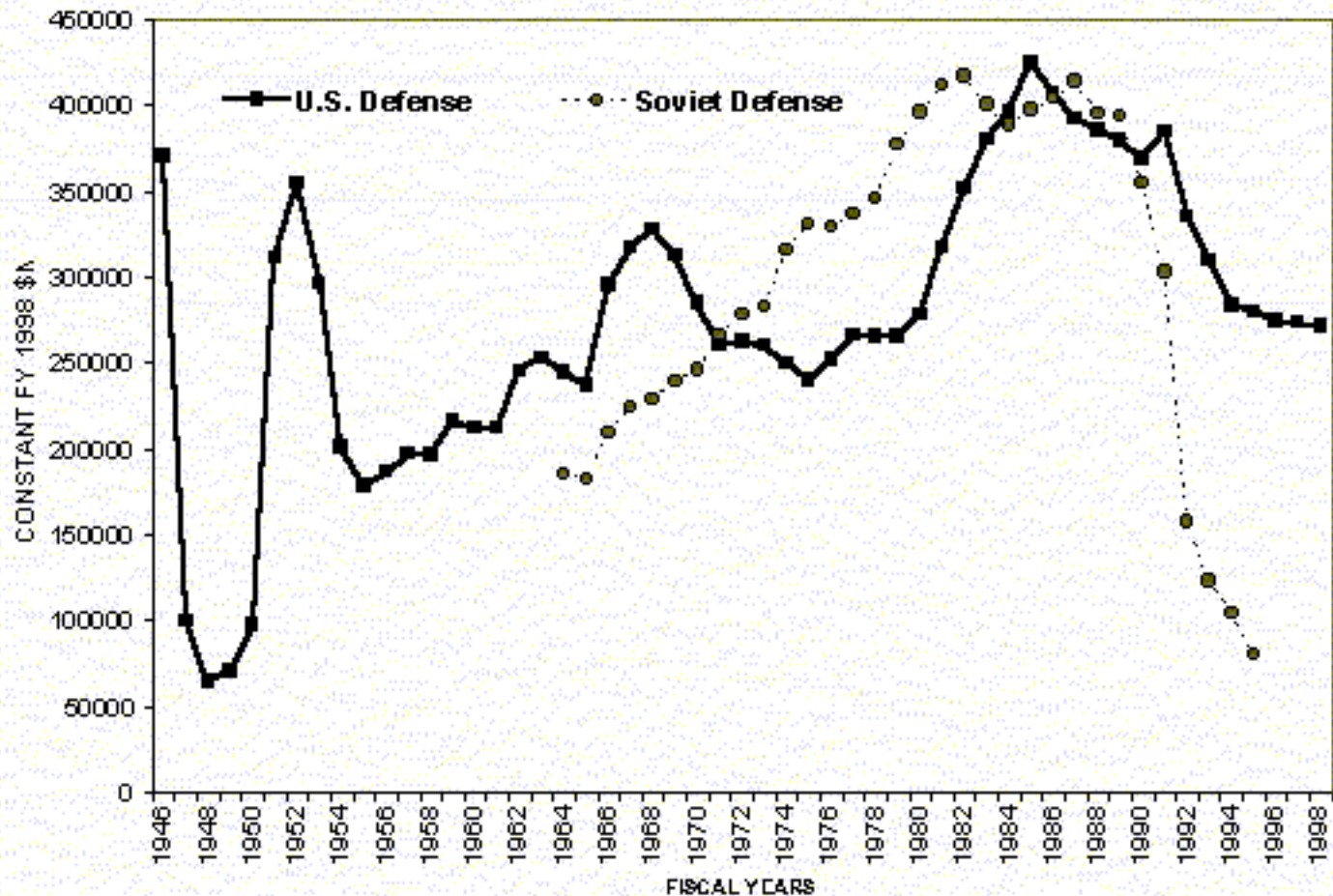
N = 530
Cesarean Section

8 weeks

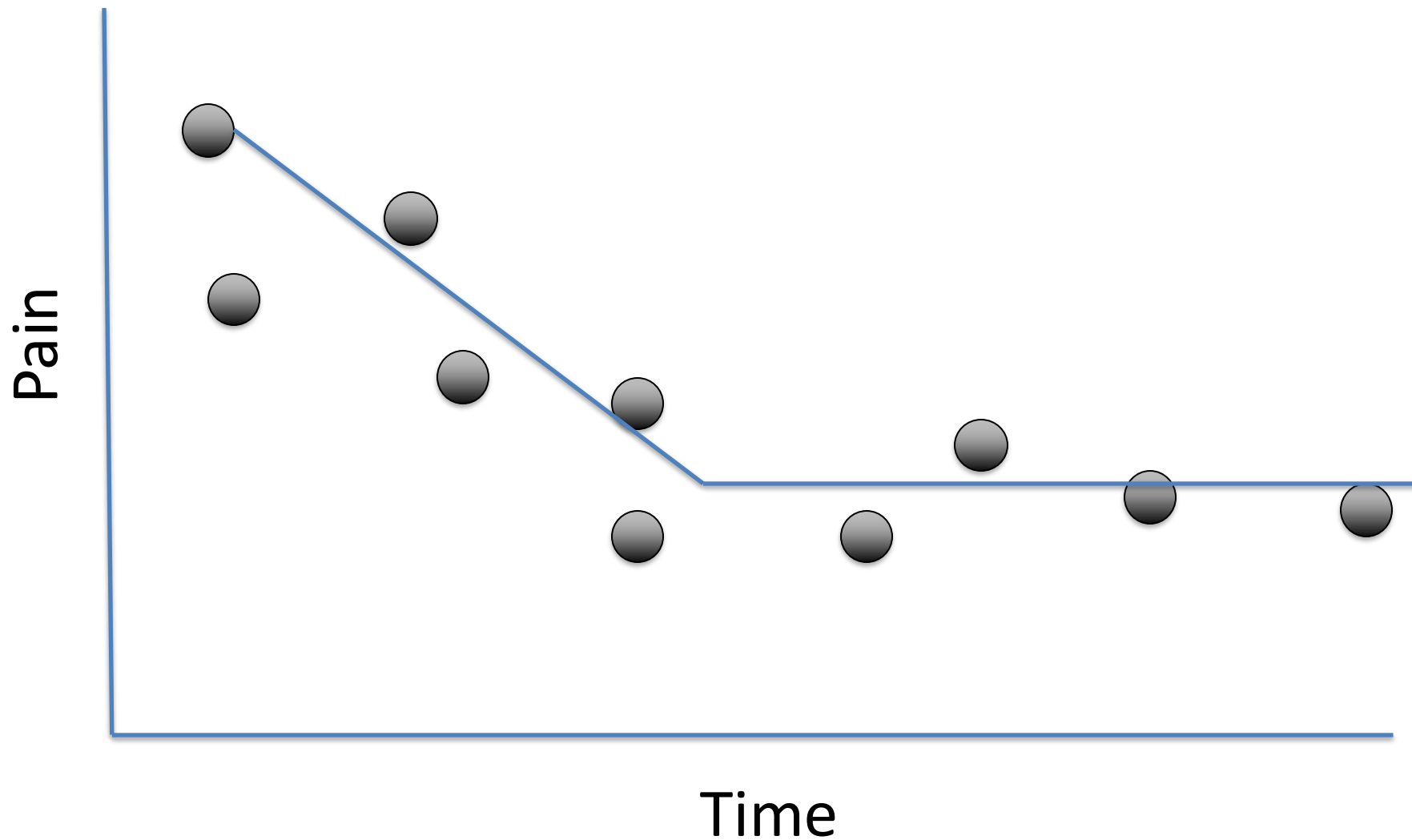
Trajectory Analysis: Model Fitting



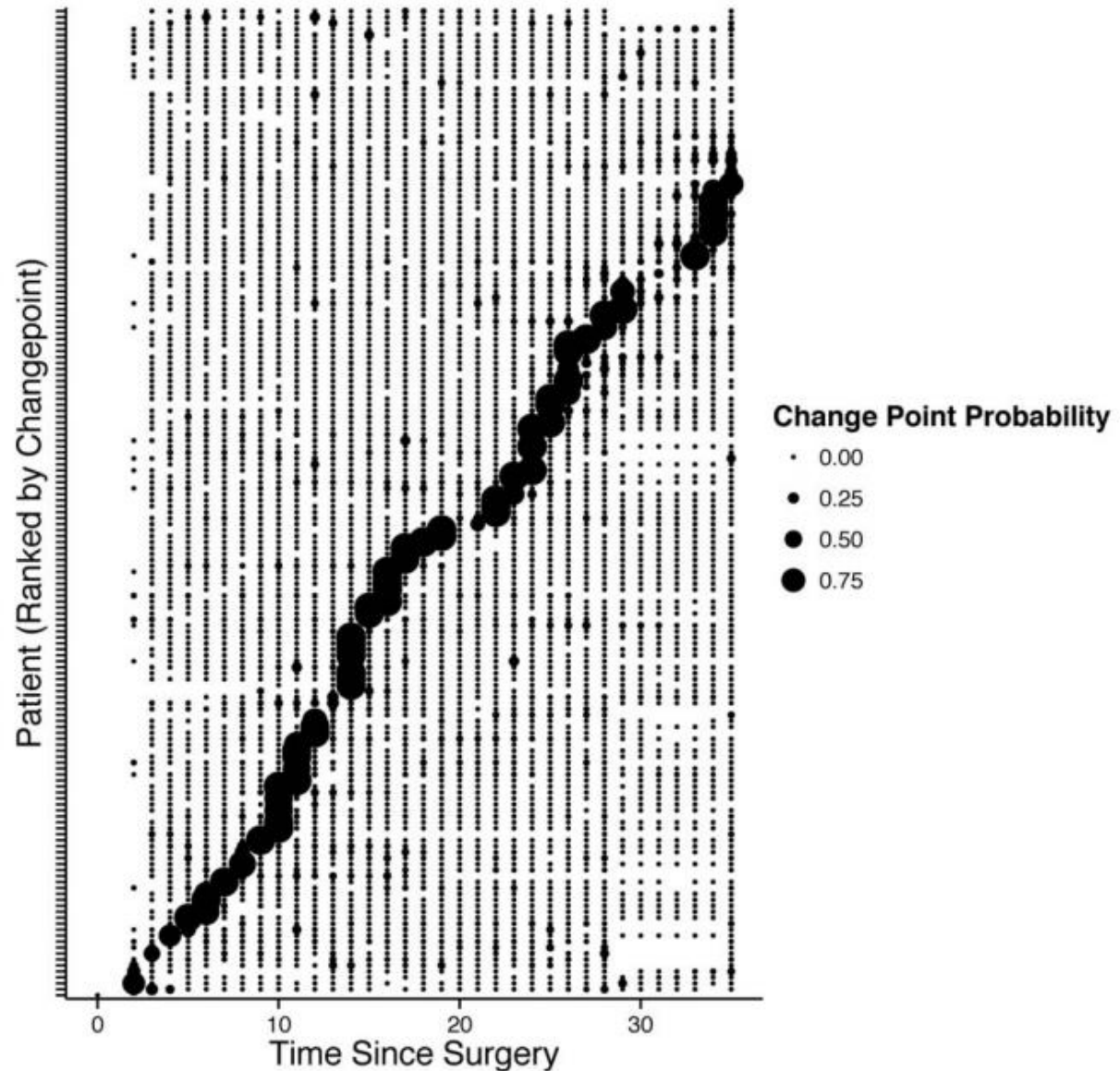
Change-Point Analysis: Soviet Military Spending



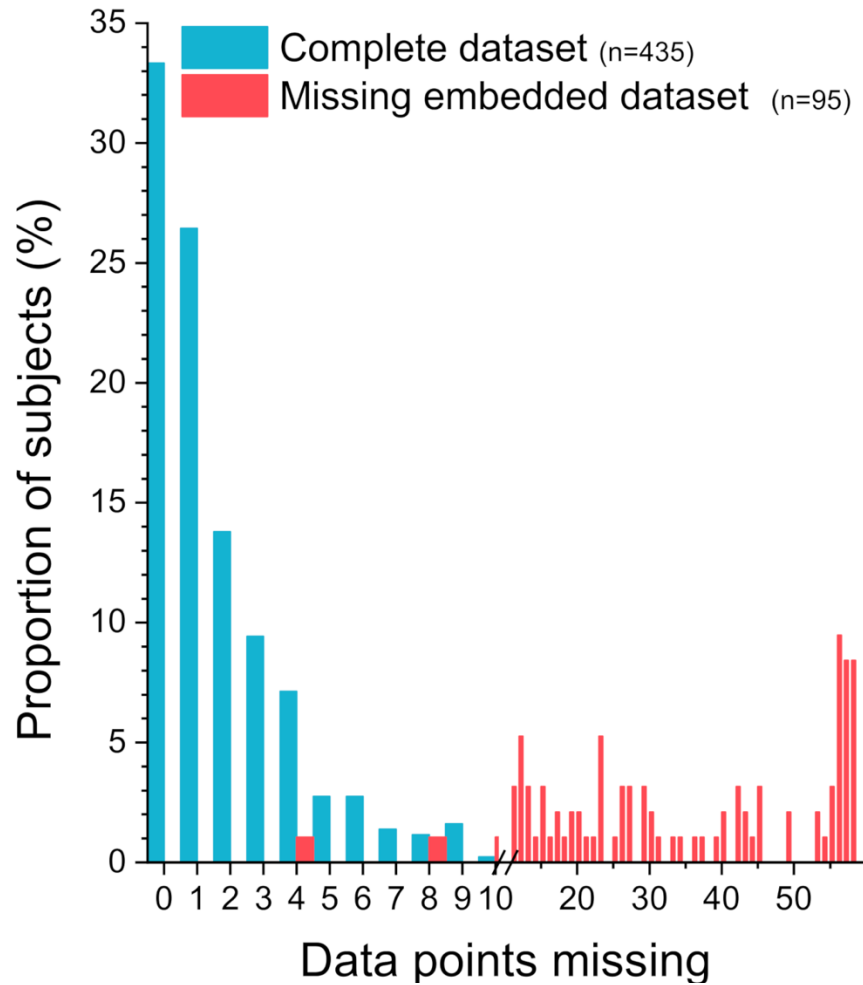
Change-Points



Change Point Locations

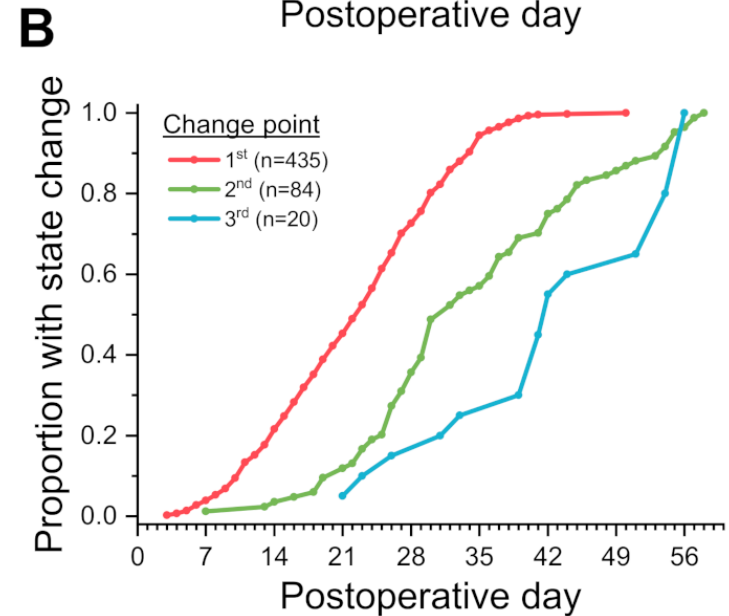
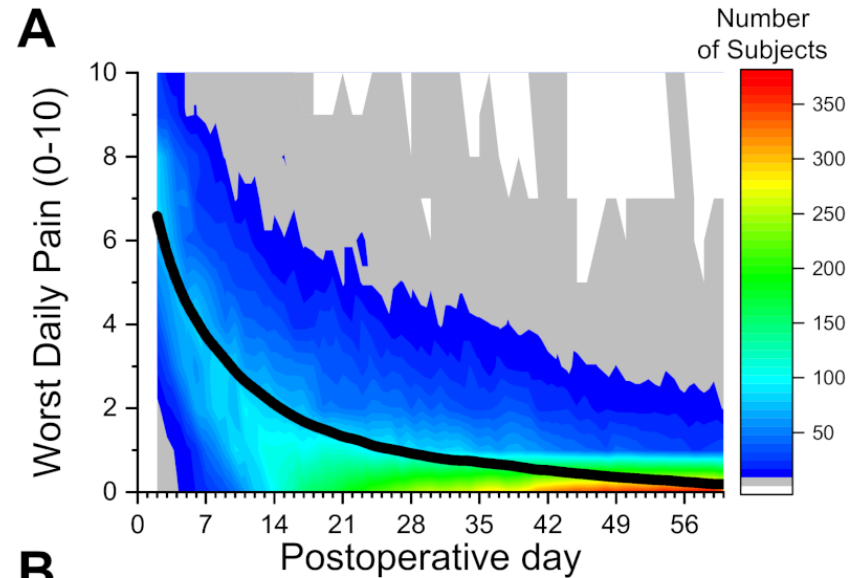


Trajectory Analysis

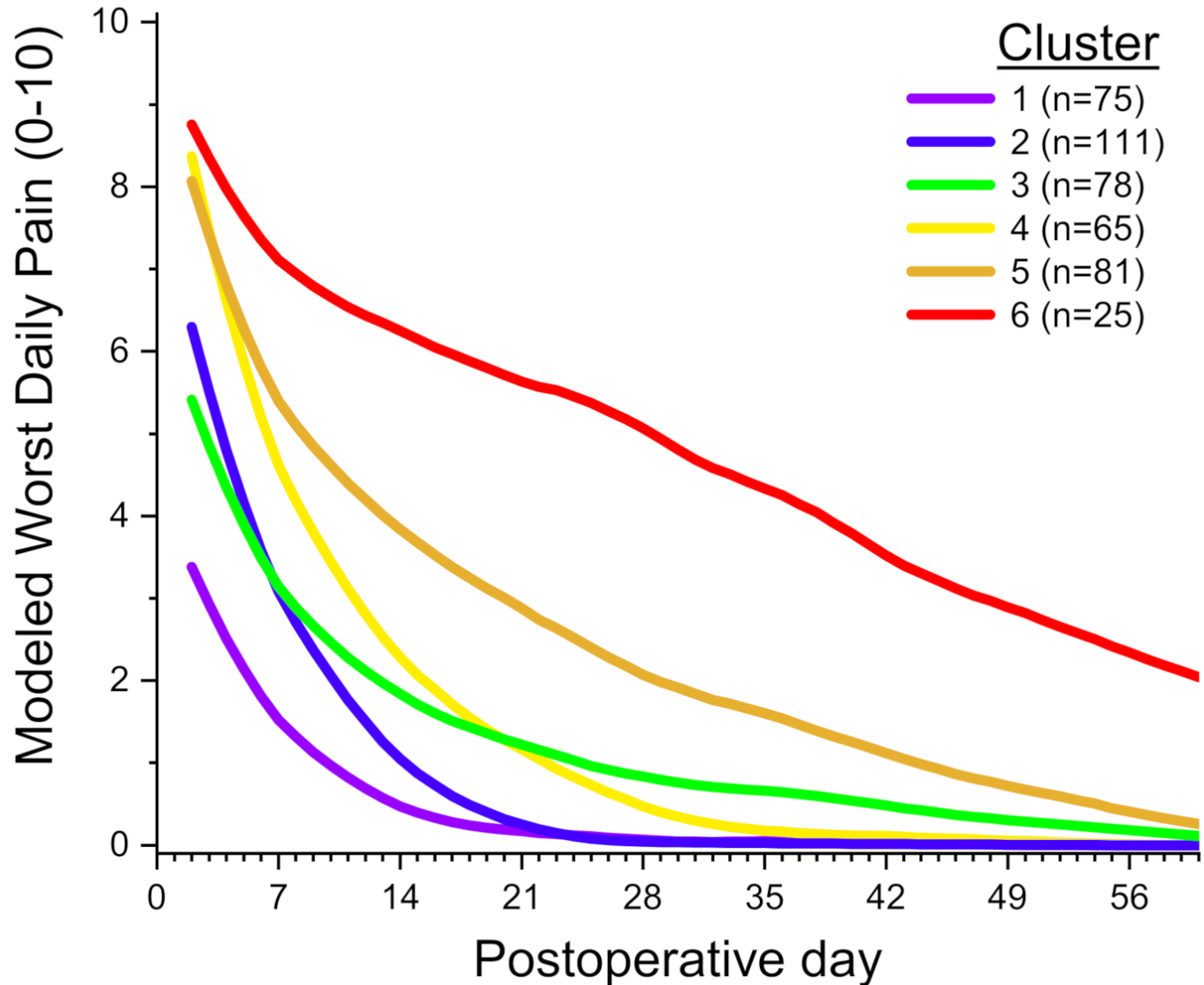


PAIN

Patterns of recovery from pain after cesarean delivery



Latent Class Trajectories



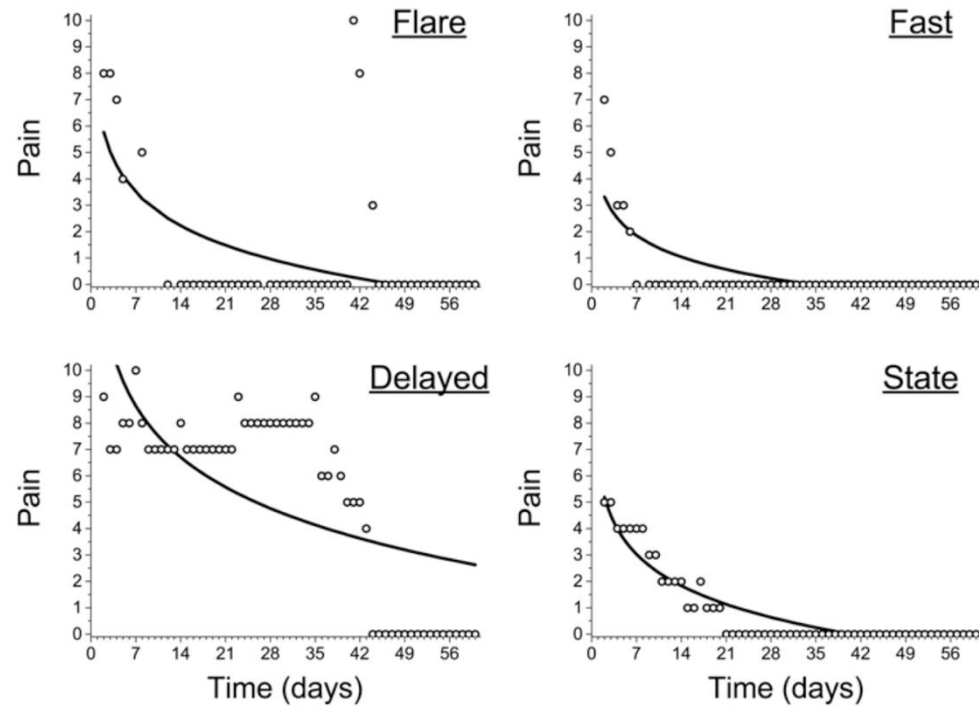
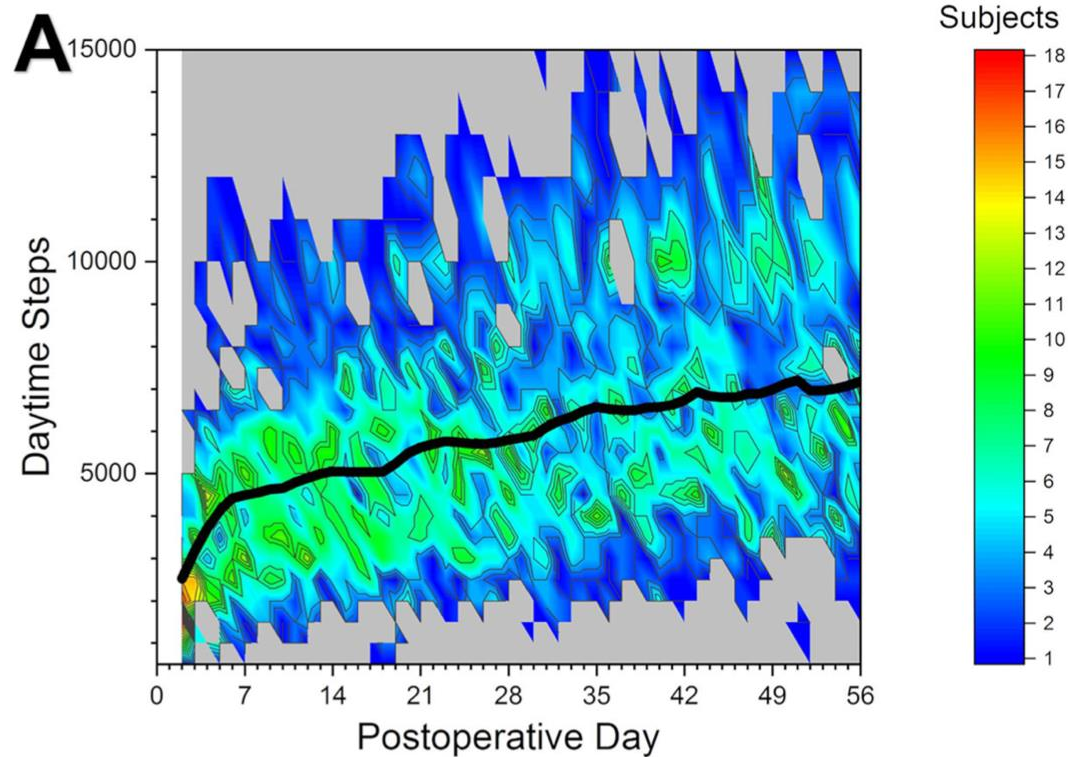


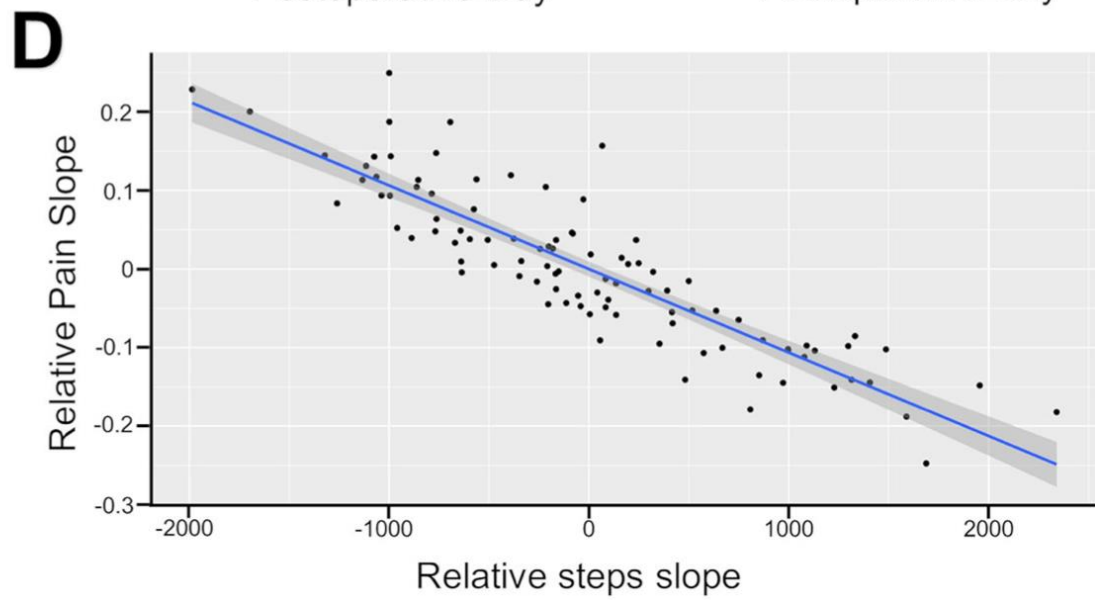
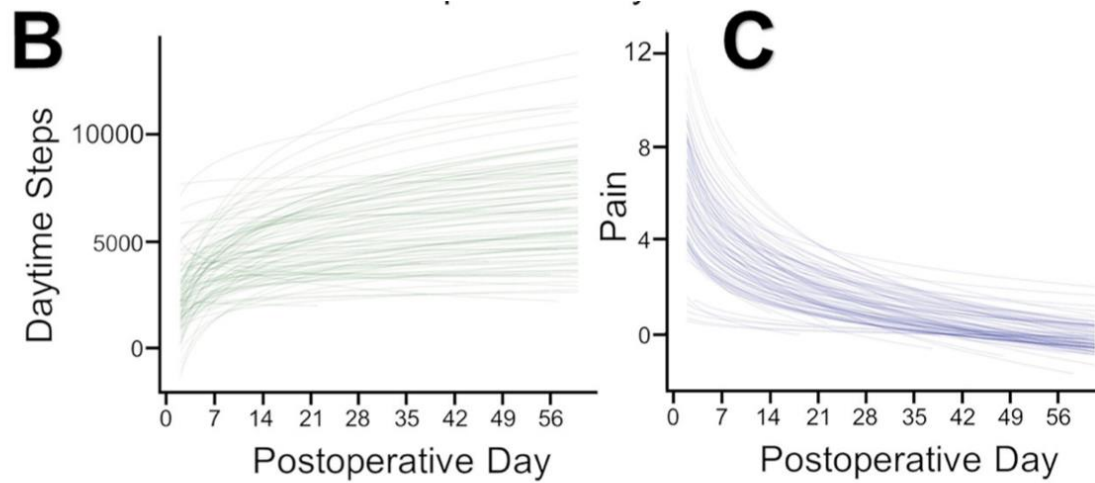
Figure 4.

Examples of change processes that induce a pattern of errors into the residuals. In each example, the distance from the prediction line to the observed point contains a temporal pattern such that points nearby can be used to predict the error of subsequent points. This can be the case because of many “0” pain scores in a sequence (“fast”), because a small group of points are very aberrant (“flare”), because the change process is not typical of a log(time model) (delayed), or because the pain scores follow a state-like pattern where the only change levels occasionally.

Recovery of physical activity after cesarean delivery and its relationship with pain

Emily E. Sharpe¹, Jessica L. Booth², Timothy T. Houle³, Peter H. Pan⁴, Lynnette C. Harris⁴, Carol A. Aschenbrenner⁴, James C. Eisenach⁴





Randomized controlled trial of intrathecal oxytocin on speed of recovery after hip arthroplasty

James C. Eisenach^{a,*}, John S. Shields^b, Robert S. Weller^a, Regina S. Curry^a, Maxwell K. Langfitt^b, Daryl S. Henshaw^a, David C. Pollock^b, Christopher J. Edwards^a, Spinal Oxytocin Hip Surgery Collaborators, Timothy T. Houle^c

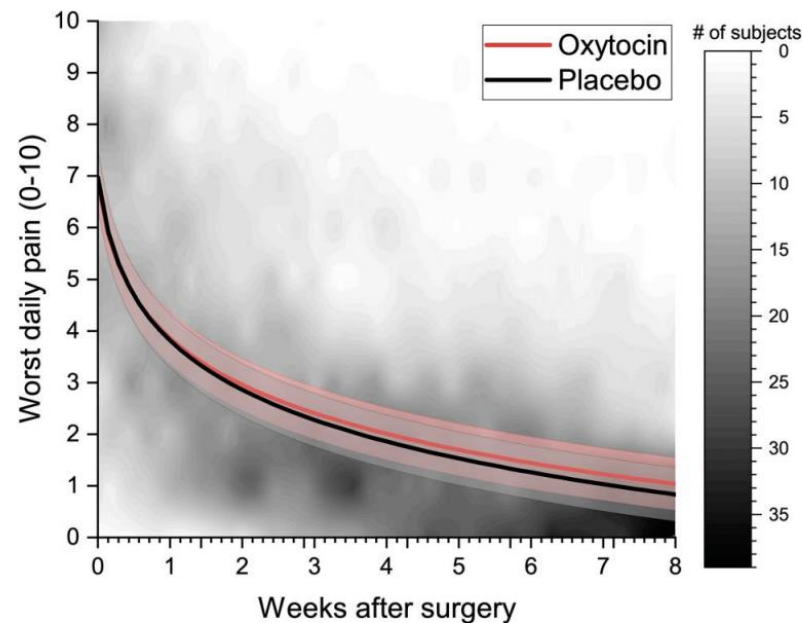
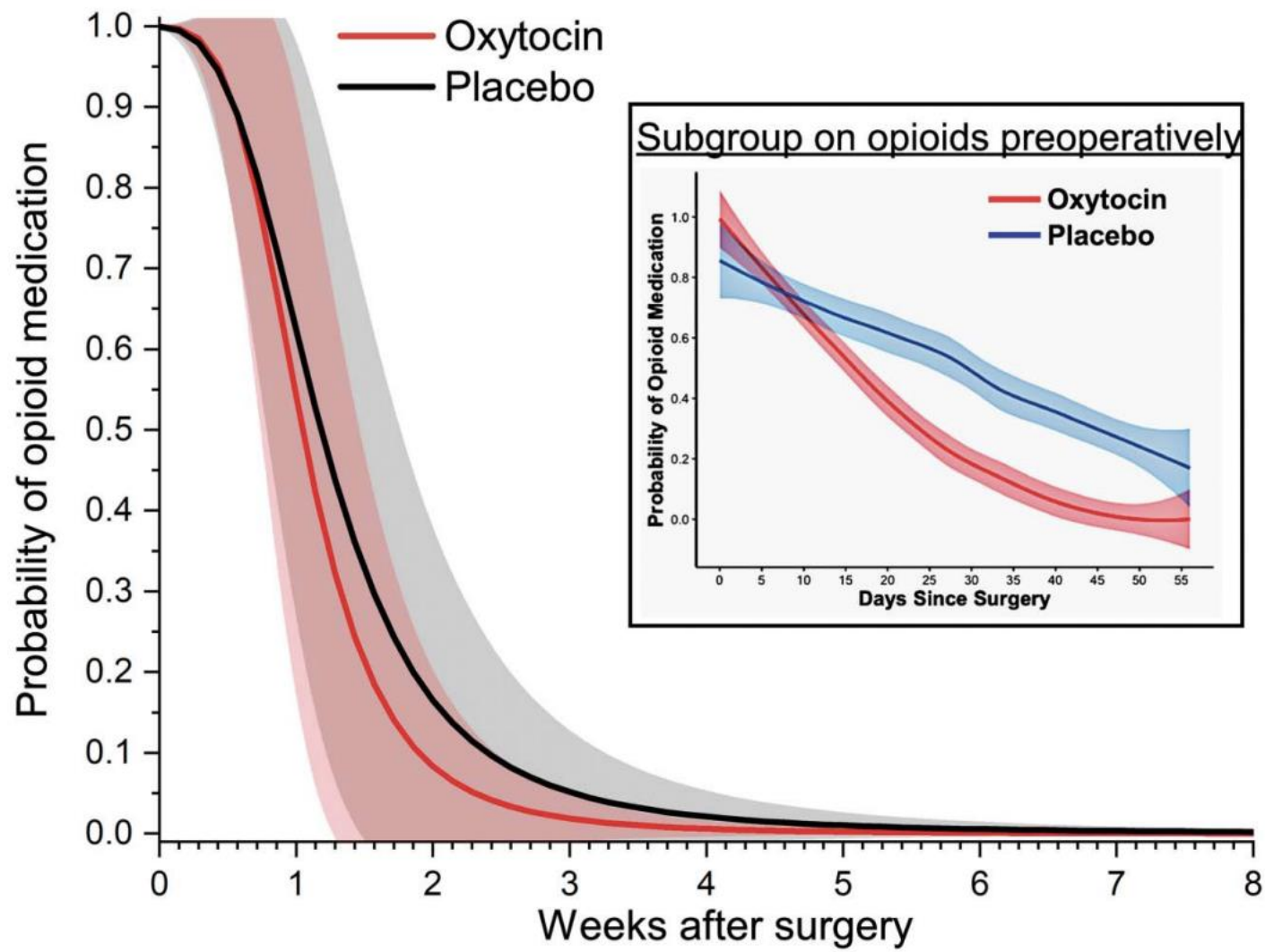
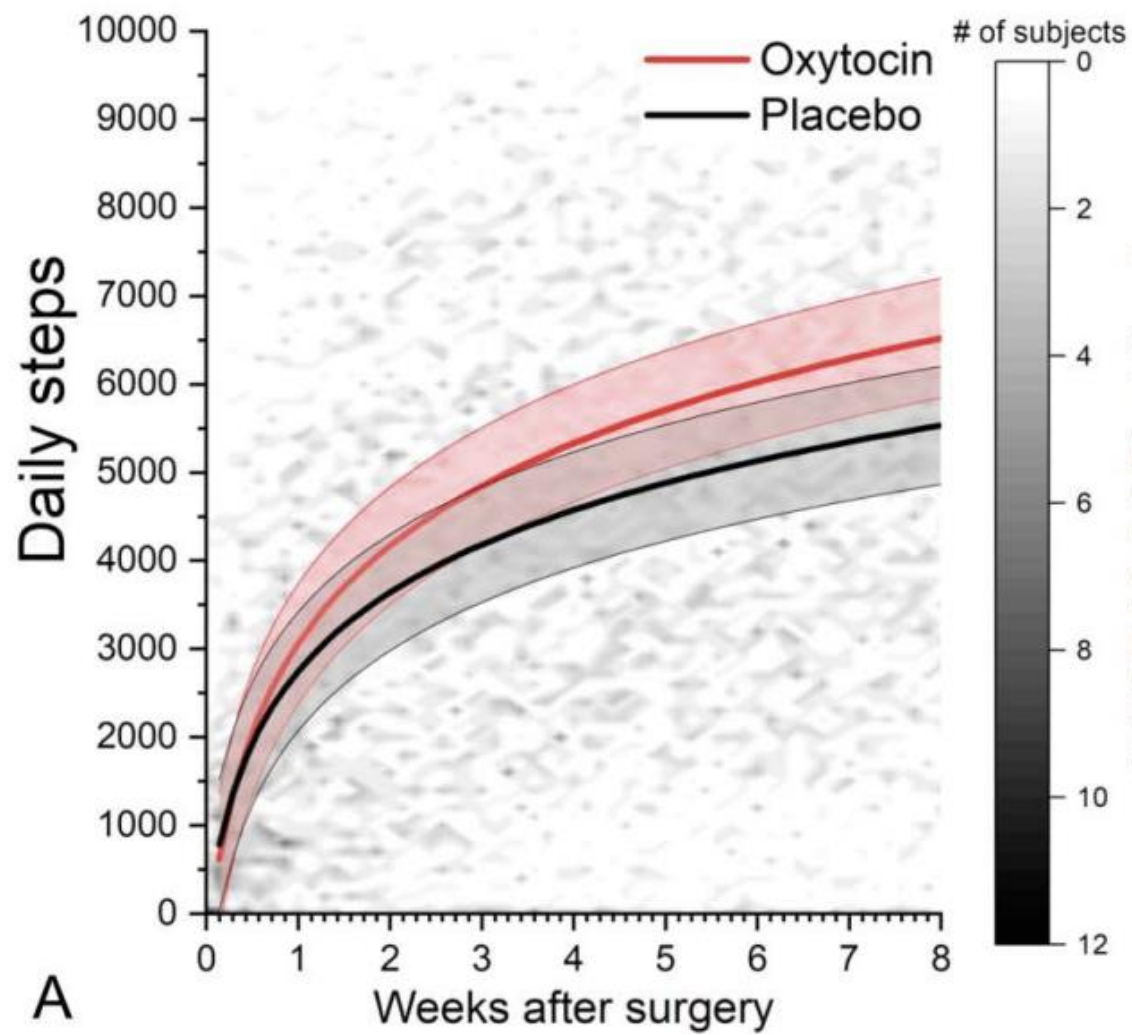


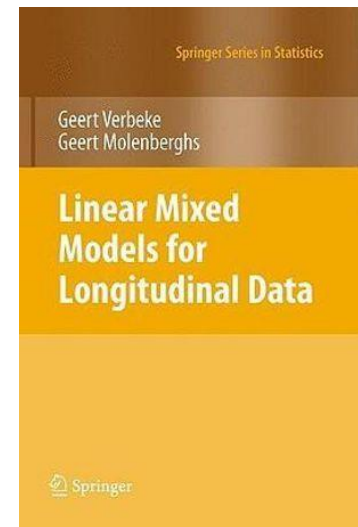
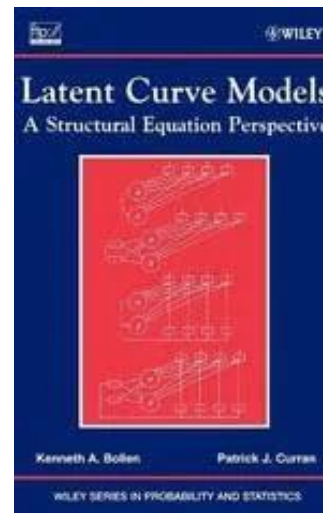
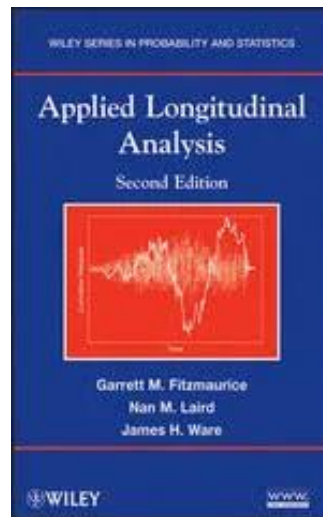
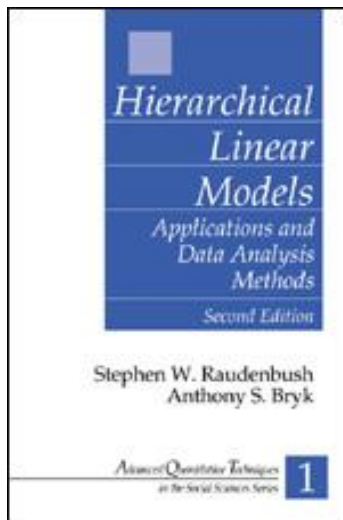
Figure 2. Primary outcome: Recovery from pain after surgery. Worst daily pain for the first 8 weeks after surgery. Lines with shaded limits represent the modelled worst daily pain score after adjustment for predefined prognostic covariates for subjects randomized to oxytocin (red) or placebo (black) with their 95% confidence limits shown in pink and grey, respectively. The background is a heat map of the raw, unmodelled, and unadjusted data showing the decreasing pain scores over time in the study population as a whole. Groups do not differ ($P = 0.057$).





Trends in Modeling Pain After Surgery Injury

- From cross-sectional designs (i.e., panel designs) to daily diary (i.e., time-series or interrupted time-series)
- From 'mere' hypothesis testing to modeling (e.g., growth curve models)
- From specifying and estimating a model on the same sample to external validation samples
- Recognition of individual-differences using random effects (intercepts and slopes as-outcomes models)
- 'Drawing strength' from previous samples using Bayesian methods



Thank You!

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Collaborators:

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